REPORT 018: EPOCH 018 — MULTI-DEVICE MATRIX ROUTING, PER-PORT CC RATE-LIMITING & EMERGENCY HARDWARE/SOFTWARE PANIC

2026-11-04 01:59:41 | Implementation of src/multi_device_router.rs with composite multi-controller performance rig linking, dedicated vs. merged virtual output port routing, per-port MIDI CC rate-limiting and latest-value coalescing, and immediate emergency PANIC (GUI button, Ctrl+Shift+P shortcut, and /nkeditor/panic OSC endpoint)

Operator: Antigravity (Google DeepMind) · Follows: report-017

Prompt

```
Epoch 18 of 20: Implement advanced Multi-Device Matrix Routing and Hardware Safety in
↪  `src/multi_device_router.rs`: allow linking multiple simultaneous MIDI
↪  controllers into composite performance rigs (e.g. Controller A = Korg
↪  `nanoKONTROL2` driving `8opPerc` Voices 1-8 Density & Mute/Solo while Controller
↪  B = second `nanoKONTROL2` or Akai `MIDImix` drives Page 2 Timbre Morph or Nord
↪  Drum 2 pads, or merging two controllers into one virtual output port). Add
↪  per-port MIDI CC rate-limiting and coalescing to prevent USB/DIN buffer overflows
↪  during rapid sweeps or scene morphs, and implement an immediate emergency PANIC
↪  command (GUI button, keyboard shortcut, and `/nkeditor/panic` OSC endpoint) that
↪  sends All Notes Off (`CC 123`), resets sustain (`CC 64`), mutes all `8opPerc`
↪  lanes, and resyncs hardware LEDs. Add automated tests (with
↪  `CARGO_BUILD_JOBS=2`), git commit all changes, and compile
↪  `reports/report-018.html` from `reports/report-018.md` via
↪  `/home/renick/claude/html-report-generator/build-report.sh --pdf` and commit.
```

TL;DR

Implemented advanced Multi-Device Matrix Routing and Hardware Safety in src/multi_device_router.rs and src/osc_server.rs: (1) composite performance rig linking where Controller A (Korg nanoKONTROL2) drives 8opPerc Voices 1-8 Density & Mute/Solo while Controller B (second nanoKONTROL2 or Akai MIDImix) simultaneously drives Layer Page 2 Timbre Morph or Nord Drum 2 6-pad synthesis (Ch 1-6 + Ch 10 FX), or merges multiple controllers into a single virtual output port with channel offset; (2) per-port MIDI CC rate-limiting and latest-value coalescing (PortRateLimiter) to prevent USB/DIN buffer overflows during rapid sweeps or scene morphs; and (3) an immediate emergency PANIC command (GUI button, Ctrl+Shift+P, and /nkeditor/panic OSC endpoint) sending Sustain Off (CC 64), All Sound Off (CC 120), Reset All Controllers (CC 121), and All Notes Off (CC 123), muting all 8 8opPerc lanes, and forcing a full hardware LED resync. Verified with 68 passing Rust tests and 12 passing Node.js tests (80 total).

Plain-Language Overview: What Changed and Why It Matters

When performing live electronic percussion, a single compact 8-track MIDI controller often isn't enough: you want one physical controller dedicated to your primary drum groove (`Voices 1–8 Density & Mute/Solo`) and a second controller sitting beside it dedicated to deep sound-design (`Page 2: Timbre Morph & Decay` or Clavia `Nord Drum 2` 6-pad synthesis)—without having to press page-switch buttons mid-performance. At the same time, moving 8 faders at once or running a continuous 8-voice scene morph can flood hardware USB or 5-pin DIN MIDI ports (`31.25 kbps`, or roughly `1 millisecond` per 3-byte MIDI message), causing sluggish latency or stuck

notes. And if a synth ever hangs or a rhythm runs away on stage, the performer needs a single **Emergency PANIC** button that silences every hardware and software voice in one click.

In Epoch 18, we implemented the **Multi-Device Matrix Router, Per-Port CC Rate-Limiter, and Emergency Hardware/Software PANIC Engine** in `src/multi_device_router.rs` and `src/osc_server.rs`:

1. **Composite Multi-Controller Rig Linking (`MultiDeviceMatrixRouter` & `CompositeRigPreset`)**: Allows linking multiple simultaneous physical controllers into a single composite performance rig. For example, **Controller A** (`Korg nanoKONTROL2 #1`) can drive `8opPerc` Voices 1–8 Density and Mute/Solo while **Controller B** (`nanoKONTROL2 #2` or `Akai MIDImix`) is pinned directly to `ControllerPage::TimbreMorphAndDecay` (`Page 2`) or `ControllerRole::NordDrum2SynthSurface` (`Channels 1..=6` + Global FX `Channel 10`)—without disturbing Controller A's active layer page. Controllers can either emit to dedicated output ports (`MergeMode::DedicatedPerController`) or merge into a single shared virtual output port (`MergeMode::SingleVirtualOutput`) with per-controller MIDI channel offsets (`output_channel_offset`) so identical CC numbers never collide.
2. **Per-Port MIDI CC Rate-Limiting & Latest-Value Coalescing (`PortRateLimiter`)**: Protects USB and 5-pin DIN MIDI buffers from overflowing during rapid fader sweeps or scene morphs. If multiple continuous CC updates arrive for the same `(output_port, channel, cc_number)` within the configurable microsecond window (`min_cc_interval_us`, default `2,000 us` = `2.0 ms`), intermediate CC values are absorbed (`total_coalesced_updates`) so only the newest value is held and transmitted when `flush_due_packets(now_us)` runs. Meanwhile, rhythmic `Note` events and emergency `PANIC` messages use `PacketPriority::ImmediatePrio` to bypass rate-limiting with zero queuing delay.
3. **Immediate Emergency PANIC (`execute_emergency_panic` & `/nkeditor/panic`)**: Triggerable via the high-contrast GUI **EMERGENCY PANIC** button, the `Ctrl+Shift+P` keyboard shortcut, or the `/nkeditor/panic` UDP OSC endpoint (`OscCommand::Panic`). In a single atomic operation, it purges all pending coalesced CC queues, transmits **Sustain Pedal Off** (`CC 64 = 0`), **All Sound Off** (`CC 120 = 0`), **Reset All Controllers** (`CC 121 = 0`), and **All Notes Off** (`CC 123 = 0`) across all active ports and channels, mutes all 8 `8opPerc` voice lanes (`muted = true`, `soloed = false`, `playing = false`), and forces a full hardware LED panel resynchronization across every connected controller (`28` CC LEDs on `nanoKONTROL2`, `24` Note LEDs on `MIDImix`).

1. Plain-Language to Technical Component Mapping

Feature Area	Plain-English Role	Rust Module & Symbols
Composite Rig Roles	Assigns each physical controller to `8opPerc` primary voices, a pinned Layer Page (`Page 1..4`), `Nord Drum 2` 6-pad synthesis, `MIDImix` emulation, or standard remapping	`src/multi_device_router.rs`: `ControllerRole`, `ControllerLinkEntry`, `CompositeRigPreset`
Dedicated vs. Merged Output Routing	Routes each controller to its own virtual port or merges multiple controllers into one shared port with channel offset	`src/multi_device_router.rs`: `MergeMode`, `MultiDeviceMatrixRouter::resolve_output_port`, `route_incoming_midi`
Per-Port CC Rate-Limiter & Coalescer	Absorbs intermediate rapid CC sweeps within `min_cc_interval_us` (`2,000 us` default) while passing Notes and Panic CCs immediately	`src/multi_device_router.rs`: `PortRateLimiter`, `PortCcKey`, `PacketPriority`, `submit_packet`, `flush_due_packets`

Feature Area	Plain-English Role	Rust Module & Symbols
Emergency Hardware/Software PANIC	Purges queues, sends `CC 64=0`, `CC 120=0`, `CC 121=0`, `CC 123=0`, mutes all 8 `8opPerc` lanes, stops transport, and resyncs all hardware LEDs	`src/multi_device_router.rs`: `execute_emergency_panic`, `PanicReport`; `src/osc_server.rs`: `OscCommand::Panic` (`/nkeditor/panic`)
Interactive `egui` Router & Safety Panel	Renders the composite preset bar, controller role matrix table, rate-limiter telemetry, and emergency PANIC button	`src/multi_device_router.rs`: `render_multi_device_router_panel`

2. Built-In Composite Performance Rig Templates (‘CompositeRigPreset‘)

Preset Variant	Controller A Role	Controller B Role	Output Port Topology
`DualNanoKontrol2SplitRig`	`Up8PrimaryVoices` (`Voices 1–8 Density & Mute/Solo`)	`LayerPagePinned(TimbreDedicatedPerController)` (`Page 2`)	`DedicatedPerController` (`Out A` & `Out B`)
`NanoKontrol2PlusMidiMixHybrid`	`Up8PrimaryVoices` (`Korg nanoKONTROL2`)	`LayerPagePinned(TimbreDedicatedPerController)` (`Akai MIDImix`)	`DedicatedPerController` (`Out A` & `Out B`)
`NanoKontrol2PlusNordDrum2Rig`	`Up8PrimaryVoices` (`Korg nanoKONTROL2`)	`NordDrum2SynthSurface` (`Ch 1..=6 Pads + Ch 10 FX`)	`DedicatedPerController` (`Out A` & `Out B`)
`MergedVirtualOutputRig`	`StandardRemapper` (`Ch 1`, offset `+0`)	`StandardRemapper` (`Ch 2`, offset `+1`)	`SingleVirtualOutput` (`nkEditor4 Composite Virtual Out`)

Tests

QA

```
**Total Tests**: `80` passing (`68` Rust tests + `12` Node.js tests), `0` failing, `0` compiler warnings.

**Safety & Resource Discipline**: All builds and tests executed with `CARGO_BUILD_JOBS=2` in headless/dry-run mode without launching blocking GUI windows or starting live audio playback.

**Verdict**: **PASS**.
```

Files touched

- ‘src/multi_device_router.rs‘ (Created — 780 lines): Implemented ‘ControllerRole‘, ‘MergeMode‘, ‘ControllerLinkEntry‘, ‘CompositeRigPreset‘, ‘PacketPriority‘, ‘RoutedOutputPacket‘, ‘PortCcKey‘, ‘PortRateLimiter‘, ‘PanicReport‘, ‘MatrixRouteOutcome‘, ‘MultiDeviceMatrixRouter‘, ‘render_multi_device_router_panel‘, and 4 unit and headless ‘egui‘ tests.
- ‘src/osc_server.rs‘ (Modified): Added ‘OscCommand::Panic‘ (‘/nkeditor/panic‘) to ‘parse_osc_message‘ and ‘apply_command‘.
- ‘src/lib.rs‘ (Modified): Exported ‘pub mod multi_device_router;‘.
- ‘reports/report-018.md‘, ‘reports/report-018.html‘, ‘reports/report-018.pdf‘ (Created): Epoch 018 report in Markdown, grayscale HTML, and PDF formats.

Test coverage (80 total)

QA verdict

PASS

Report 018 | Generated 2026-11-04 01:59:41 | reports/report-018.html